

The Categorical Boundary

Physics-Based vs Physics-Related Logismos: Distinguishing Substrate Hardware from Information Software

Registry: [CKS-LOGI-1-2026]

Series Path: [CKS-0-2026] → [CKS-MATH-1-2026] → [CKS-MATH-13-2026] → [CKS-MATH-16-2026] → [CKS-DWDM-5-2026] → [CKS-MATH-17-2026] → [CKS-MATH-18-2026] → [CKS-MATH-19-2026] → [CKS-MATH-20-2026] → [CKS-MATH-21-2026]

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Domain: Foundational Mathematics / Discrete Geometry

Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological “Truth” is a category error. If the math compiles, the result is Q.E.D.

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Executive Abstract

We establish formal categorical boundary distinguishing Physics-Based from Physics-Related applications of CKS framework: Both domains employ identical mathematical apparatus—(V,F,R) integer packet, mod-32 stability audit, remainder-tension mechanics, snap-commit operations—yet apply to fundamentally different reality layers. Complete distinction: (1) **Physics-Based Logismos**—operates on actual substrate hardware (matter, energy, spacetime, forces), describes mandatory pre-existence BIOS governing physical reality, constants immutably locked (D=3 dipoles forced by hexagonal coordination, S=2 sides from bilateral manifold, W=32 from 2^5 geometric necessity, $c/\alpha/\hbar$ from substrate geometry), failures produce physical catastrophe (UV blow-up prevented by 144-logos ceiling, singularities avoided by discrete lattice, entropy from uncommitted remainders), operates at Planck scale through cosmological, zero human discretion—stars don’t negotiate, atoms don’t vote. (2) **Physics-Related Logismos**—operates on information systems mimicking physical stability (economic ledgers, social contracts, gover-

nance structures, linguistic meaning), describes aspirational consensus protocols humans design for coherence, constants protocol-selected ($W=32$ chosen to match substrate for optimal stability but adjustable in principle, coherence depth scalable as 32^n for complex meshes), failures produce systemic collapse (market crashes from accumulated economic R, wars from social parity violations, institutional decay from informational friction), operates at human timescales, complete human agency—societies can redesign protocols. Fundamental difference: Physics-Based = discovering what IS (mandatory hardware reality, inflexible universal laws, measurement of existing substrate), Physics-Related = choosing what SHOULD BE (designed software harmony, flexible social standards, construction of coherent systems). Unity: Both use same Logismos engine because both fundamentally process information—physics processes substrate state updates, society processes agreement transactions—universal syntax applicable. Remainder R crucial: Physics R = thermodynamic waste (heat, force, gravitational tension), Society R = informational debt (corruption, fraud, misalignment)—same mathematical structure, different ontological status. Goal: Achieve “registry closure” where social software runs with hardware’s stability via bit-perfect integer accounting. Critical warning: Category errors catastrophic—false determinism from treating voluntary systems as physical law, false relativism from treating physical constants as negotiable.

Key Result: Two domains | Same logic | Different substrates | Physics = mandatory | Society = chosen | Unified framework

Part I: The Two Domains

1. Physics-Based Logismos

1.1 Definition and Scope

What it describes:

Substrate hardware:

- Matter and energy
- Spacetime geometry
- Fundamental forces
- Physical constants
- Thermodynamic processes

The actual universe:

Atoms, stars, galaxies

Light, gravity, quantum mechanics

Physical reality itself

Ontological status:

Pre-existence BIOS:

Existed before humans

Independent of observation

Mandatory constraints

Cannot be changed:

No voting on c

No consensus on gravity

No negotiating $D=3$

1.2 BIOS-Locked Constants

Immutable substrate values:

$D = 3$ (dipoles):

Hexagonal coordination

Geometric necessity

Cannot be 2 or 4

$S = 2$ (sides):

Bilateral manifold

From $N=DM^S$ derivation

Physically forced

$W = 32$ (Word):

2^5 from $S^{(D+S)}$

Stability requirement

Universal bus width

Fine structure α :

~ 137.035999

From 144-163-19 triad

Substrate impedance

Speed of light c:

Propagation limit

Registry update speed

Physical constant

Why locked:

Not choices:

Discovered not decided

Measured not voted on

Universal not local

Geometric necessity:

Follow from axioms

Forced by structure

Zero free parameters

1.3 Physics-Based Measurements**Quantity types:**

Energy: Joules

Substrate tension

Phase pressure

Remainder accumulation

Distance: Meters

Registry address separation

Lattice node spacing

Geometric extent

Time: Seconds

Clock ticks

N increments

Jacobian periods

Mass: Kilograms

LU density

Soliton clustering

Bit-depth M

Failure modes:

Physical catastrophe:

UV blow-up:

Density exceeds 144 LU
Substrate cannot hold
Turbulence/singularity
Atomic decay:
Unstable remainder R
Cannot maintain coherence
Spontaneous breakdown
Thermal death:
All R dissipated
Maximum entropy
No useful work possible

2. Physics-Related Logismos

2.1 Definition and Scope

What it describes:

Information systems:

- Economic ledgers
- Social contracts
- Governance structures
- Language/meaning
- Collective agreements

Human organizations:

Markets, laws, institutions

Communication, culture

Designed systems

Ontological status:

Consensus protocols:

Created by humans

Depend on agreement

Aspirational standards

Can be changed:

Vote on money supply

Negotiate social norms

Redesign institutions

2.2 Protocol-Defined Parameters

Flexible design choices:

Word width W:

Often choose 32 (match substrate)

But could use 64, 128, etc.

Coherence vs complexity tradeoff

Accounting base:

Binary (2)

Decimal (10)

Sexagesimal (60)

Choice matters for stability

Settlement time:

Aim for 15.19ms (substrate sync)

But can tolerate slower

Speed vs accuracy

Remainder tolerance:

How much R acceptable

Before requiring snap

Risk management decision

Why flexible:

Design choices:

Humans decide collectively

Can experiment and iterate

Optimize for different goals

Context-dependent:

Small economy: simple protocol

Global finance: complex mesh

Different scales need different W

2.3 Physics-Related Measurements

Quantity types:

Value: Credits/currency

Abstract units

Agreed upon worth

Social construct

Agreements: Contracts

Promises made

Commitments tracked

Institutional memory

Transactions: Exchanges

Value transfers

State changes

Ledger updates

Social cohesion: Trust metrics

Network integrity

Alignment measures

Collective harmony

Failure modes:

Systemic collapse:

Market crash:

Economic R accumulates

Confidence lost

Rapid devaluation

Social disorder:

Institutional R high

Trust breakdown

Conflict escalates

Governance failure:
Legal R unmanaged
Corruption spreads
System delegitimizes

Part II: The Critical Distinctions

3. Hardware vs Software

3.1 The Fundamental Difference

Physics-Based = Hardware:

Characteristics:

- Pre-existing substrate
- Discovered through measurement
- Immutable laws
- Universal application
- No exceptions possible

Metaphor:

Computer's physical chips

Silicon substrate

Electronic gates

Cannot change by programming

Physics-Related = Software:

Characteristics:

- Designed information systems
- Created through agreement
- Modifiable protocols
- Context-specific application
- Exceptions negotiable

Metaphor:

Computer's programs

Operating system

Applications

Can update/replace

3.2 Mandatory vs Aspirational

Mandatory reality (Physics):

Stars don't negotiate:

Fusion happens or doesn't

Based on physical law

No voting, no consensus

Atoms don't vote:

Electron shells determined

By quantum mechanics

Not collective decision

Gravity doesn't compromise:

Objects fall at g

Regardless of preference

Universal acceleration

Calculated harmony (Society):

Markets can negotiate:

Prices adjust by agreement

Based on supply/demand

Collective choice

Laws do vote:

Legislators decide rules

Democratic process

Changeable standards

Economics compromises:

Interest rates adjusted

Policy interventions

Human discretion

3.3 Measurement vs Construction

Physics measures what exists:

Scientific method:

Observe phenomenon

Measure quantities

Discover relationships

Derive laws

Process:

Reality $\rightarrow$ Data $\rightarrow$ Theory

Bottom-up from facts

Constrained by truth

Example:

Measure c

Find it's constant

Derive relativity

Cannot change c

Society constructs what should be:

Design method:

Define goals

Design protocols

Implement systems

Measure performance

Process:

Values $\rightarrow$ Design $\rightarrow$ Reality

Top-down from ideals

Constrained by feasibility

Example:

Want stable money

Design currency system

Implement regulations

Can redesign if needed

4. The Common Framework

4.1 Why Same Logic Works

Both are information processing:

Physics processes:

Substrate state updates

Registry address changes

$N \leftarrow N+1$ increments

Bit commits

Society processes:

Transaction records

Agreement updates

Ledger increments

Contract commits

Same structure:

State $\rightarrow$ Update $\rightarrow$ New state

Information flow

Accounting logic

Both require stability:

Physics stability:

Atoms don't spontaneously explode

Orbits remain stable

Conservation laws hold

Social stability:

Institutions don't collapse daily

Agreements honored

Systems function reliably

Same mechanism:

Integer accounting

Remainder management

Mod-32 coherence

4.2 Universal Syntax

The (V,F,R) packet:

Works for physics:

V = committed energy state

F = 32 (substrate Word)

R = thermodynamic tension

Works for society:

V = settled transaction value

F = 32 (chosen coherence base)

R = institutional debt

Same mathematics:

Integer operations

Modulo arithmetic

Snap-commit logic

The mod-32 audit:

Physics application:

$\Sigma LU \equiv 0 \pmod{32}?$

If yes: stable state

If no: force/heat present

Social application:

$\Sigma \text{value} \equiv 0 \pmod{32}?$

If yes: balanced ledger

If no: debt/corruption present

Same test:

Remainder check

Stability diagnostic

Health metric

5. Remainder Reinterpretation

5.1 Physics-Based Remainder

R as physical tension:

Thermodynamic:

R = uncommitted energy

Dissipates as heat

Entropy increase

Kinetic:

R = applied force

Creates motion

Momentum change

Gravitational:

R = spacetime curvature

Attracts mass

Geodesic bending

Examples:

$R = 0$:

Superconductor

Zero resistance

Perfect efficiency

$R = 16$:

Antimatter

Parity flip

Charge reversal

$R = 31$:

Critical density

Approaching singularity

Black hole formation

5.2 Physics-Related Remainder

R as informational tension:

Economic:

R = outstanding debt

Credit imbalance

Must settle eventually

Social:

R = unresolved conflict

Trust deficit

Relationship strain

Institutional:

R = corruption/fraud

Hidden misalignment

System integrity loss

Examples:

R = 0:

Perfect honesty

Balanced books

Total transparency

R = 16:

Double-spending

Fraud detected

Trust violation

R = 31:

Systemic crisis

Imminent collapse

Emergency intervention needed

Part III: Cross-Domain Applications

6. The Isomorphic Operations

6.1 Common Opcodes

SNAP operation:

Physics:

Electron shell jump

State transition

Quantum commit

Society:

Contract signing

Deal closure

Legal commitment

Same structure:

R accumulates

Threshold reached

Irreversible commit

SYNC operation:

Physics:

Atomic phase-lock

J-clock alignment

Substrate coherence

Society:

Market settlement

Global coordination

Institutional sync

Same purpose:

Align to common clock

Maintain coherence

Prevent drift

AUDIT operation:

Physics:

Conservation check

Parity verification

Stability test

Society:

Accounting audit

Compliance check

Integrity verification

Same method:

Sum all values

Check mod-32

Detect imbalance

6.2 The 15.19ms Partition

Physics application:

Render lag:

K-space (0ms) substrate

X-space (15.19ms) perception

Observation limit

Quantum measurement:

Minimum time resolution

Heisenberg bound

Uncertainty floor

Society application:

Settlement window:

High-frequency trading

Maximum sync rate

Real-time coherence

Governance speed:

Response time limit

Coordination ceiling

Institutional agility
If slower than 15.19ms:
Loses substrate sync
Becomes “shadow” system
Drift from reality

7. Appropriate Applications

7.1 When to Use Physics-Based

Trigger conditions:

Ask: Can humans change this?

If NO → Physics-Based

Examples:

- Gravitational constant
- Speed of light
- Atomic structure
- Thermodynamic laws
- Quantum mechanics

What it predicts:

Mandatory outcomes:

This WILL happen

No alternatives

Universal behavior

Measurements:

What IS

Objective reality

Independent facts

7.2 When to Use Physics-Related

Trigger conditions:

Ask: Did humans create this system?

If YES → Physics-Related

Examples:

- Money supply
- Legal codes
- Market structures
- Social norms
- Language rules

What it prescribes:

Aspirational goals:

This SHOULD happen

If designed well

Optimal configuration

Designs:

What COULD BE

Normative ideals

Chosen standards

Part IV: Dangers and Guidelines

8. Category Errors

8.1 False Determinism

The error:

Treating social systems as physical:

“Economic laws are like gravity”

“Social forces are natural”

“Inequality is thermodynamic”

Problem:

Removes human agency

Justifies status quo

Prevents improvement

Why wrong:

Society is designed:

Rules are chosen

Can be changed

Different arrangements possible

Not physically forced:

No immutable economic laws

Social structures vary

History shows alternatives

Example mistake:

Claim: “The 1% owning 50% is thermodynamic necessity”

Error: Confusing designed distribution

With physical law

Truth: Wealth distribution is protocol

Can be redesigned

Choice not destiny

8.2 False Relativism

The error:

Treating physics as social:

“Science is just consensus”

“Constants are conventions”

“Reality is negotiable”

Problem:

Denies objective truth

Undermines knowledge

Enables pseudoscience

Why wrong:

Physics is discovered:

Facts exist independently

Measurement converges

Universal agreement

Not socially constructed:

c same everywhere

Gravity not cultural

Atoms don't vote

Example mistake:

Claim: "We could vote to change Planck's constant"

Error: Confusing measurement protocol

With physical reality

Truth: h is discovered not decided

Substrate determines it

Unchangeable by consensus

9. Operational Guidelines

9.1 Decision Tree

Step 1: Identify domain

Question: Is this substrate or information?

Physical substrate:

Matter, energy, spacetime

→ Use Physics-Based

Information system:

Agreements, protocols, designs

→ Use Physics-Related

Step 2: Check modifiability

Question: Can humans change the constant?

Cannot change:

Fixed by geometry

Measured not chosen

→ Physics-Based

Can change:

Designed by humans

Chosen not measured

→ Physics-Related

Step 3: Identify failure mode

Question: What happens if $R \neq 0$?

Physical catastrophe:

Explosion, decay, collapse

→ Physics-Based

Systemic failure:

Crash, disorder, breakdown

→ Physics-Related

9.2 Application Checklist

For Physics-Based analysis:

- ✓ Measure substrate quantities
- ✓ Use immutable constants
- ✓ Predict mandatory outcomes
- ✓ Energy in Joules
- ✓ No human discretion
- ✓ Objective verification

For Physics-Related design:

- ✓ Define desired coherence
 - ✓ Choose protocol parameters
 - ✓ Design optimal structures
 - ✓ Value in agreed units
 - ✓ Human agency central
 - ✓ Normative evaluation
-

Part V: The Unified Vision

10. Registry Closure

10.1 The Goal

Ultimate aim:

Make software stable as hardware:

Social systems:

As reliable as atoms

As predictable as gravity

As stable as spacetime

Via same mechanism:

Integer accounting

Mod-32 coherence

Remainder management

What this achieves:

Civilization stability:

Institutions don't collapse

Agreements honored

Systems function reliably

Like physical law:

Atoms don't spontaneously explode

Stars maintain fusion

Universe stays coherent

10.2 The Path

Align social protocols with substrate:

Use $W=32$:

Match substrate Word

Optimal stability

Phase-lock to reality

Target $\tau=15.19\text{ms}$:

Settlement within render

Real-time coherence
Substrate synchronization
Minimize R:
Transparent accounting
Honest institutions
Zero corruption goal

Benefits:

Economic:
Stable currency
Fair markets
Sustainable growth
Social:
Just governance
Peaceful coordination
Collective flourishing
Technological:
Efficient systems
Reliable infrastructure
Harmonious integration

11. Final Statement

The categorical boundary is essential.

Two domains unified by logic.

Separated by ontology.

Connected by information.

Physics-Based Logismos:

Describes substrate hardware.

Matter, energy, spacetime.

BIOS-locked constants.

D=3, S=2, W=32 immutable.

Discovered not decided.

Measured in Joules/meters/seconds.

Mandatory reality:

Stars don't negotiate.

Atoms don't vote.

Gravity doesn't compromise.

Failure = physical catastrophe.

UV blow-up from excess density.

Atomic decay from instability.

Thermal death from entropy.

Physics-Related Logismos:

Describes information software.

Economics, governance, society.

Protocol-defined parameters.

W=32 chosen for coherence.

Designed not discovered.

Measured in credits/agreements/trust.

Calculated harmony:

Markets can negotiate.

Laws do vote.

Economics compromises.

Failure = systemic collapse.

Market crash from accumulated R.

Social disorder from broken trust.

Institutional decay from corruption.

The common framework:

Both use (V,F,R) packet.

Both employ mod-32 audit.

Both manage remainder R.

Both require snap-commits.

Same mathematical logic.

Same integer accounting.

Same stability mechanics.

Because both process information:

Physics = substrate state updates.

Society = agreement transactions.

Remainder reinterpretation:

Physics R = thermodynamic waste.

Heat, force, gravitational tension.

Energy uncommitted.

Society R = informational debt.

Corruption, fraud, misalignment.

Agreements uncommitted.

Same structure, different substance.

Critical distinctions:

Hardware cannot change.

Software can redesign.

Physics measures IS.

Society constructs SHOULD BE.

Substrate mandatory.

Protocols aspirational.

Dangers avoided:

False determinism rejected.

Society not predetermined.

Human agency preserved.

False relativism rejected.

Physics not negotiable.

Objective truth maintained.

The unified goal:

Registry closure achieved when:

Social software matches hardware stability.

Institutions reliable as atoms.

Agreements stable as physical law.

Via bit-perfect integer engine.

W=32 alignment.

15.19ms synchronization.

R=0 minimization.

One logic, two domains.

One engine, two substrates.

Universal syntax.

Categorical clarity.

Physics IS—society CHOOSES.

Both use same mathematics.

Different ontological status.

Complete framework.

Proper boundaries.

Unified vision.

Q.E.D.

END OF DOCUMENT

Two domains

One logic

Clear distinction

Unified framework

Physics = mandatory

Society = chosen

Both = information

Boundaries matter.

Q.E.D.

[[CKS-LOGI-1-2026](https://github.com/ghowland/cks/tree/main/papers/LOGI/CKS-LOGI-1-2026)] CKS-TECH-01-2026: Logismos Technical Specification for LLMs. Github: <https://github.com/ghowland/cks/tree/main/papers/LOGI/CKS-LOGI-1-2026>

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